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Certified benchmarking of discrete IS 800:2007-compliant space-truss sizing: an exact MILP baseline and a GA–local-search hybrid with local-optimality certificates

Short title: Certified benchmarking of discrete IS 800 truss sizing

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Data availability. All benchmark definitions, the SP 6(1) catalogue, the raw results of every run (JSON), the MILP certificates and the scripts that regenerate every table and figure are openly available at `https://github.com/mdaipayan/GA-MINLP-IS800-Truss-Optimization` (folders `src_v2`, `results_v2`, `figures_v2`) and archived on Zenodo, `https://doi.org/10.5281/zenodo.20508136` [update to the DOI of the v2 release].

Code availability. The Python code (analysis engine, exact MILP, all optimizers, statistics) is released under the licence stated in the repository above.

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Abstract

Metaheuristic studies of discrete truss sizing usually report the lightest design they found, but they rarely show how far this design is from the true optimum, and they often compare methods with different numbers of structural analyses. This paper addresses both problems for steel space trusses designed to IS 800:2007 with SP 6(1) equal-angle sections. First, following earlier exact formulations for trusses and for Eurocode frames, we write the complete IS 800 problem—tension capacity, the column-buckling curve, slenderness and displacement limits—as an exact mixed-integer linear program (MILP), and use it as a yardstick for the heuristics. Solved by branch-and-bound certifies the global optimum of the 6-bar (101.17 kg) and the 25-bar (580.06 kg) benchmarks, and gives a proved lower bound for the 72-bar benchmark. Second, we propose GA–DNS, a genetic algorithm followed by a deterministic discrete neighbourhood search that ends with a local-optimality certificate. GA–DNS and six metaheuristics (SA, GA, PSO, ACO, BB-BC, DE) are compared over 20 seeded runs with the same analysis budget and the same feasibility rule. GA–DNS reached the certified optimum in 95 % (6-bar) and 90 % (25-bar) of runs, and its final design was certified locally optimal in 59 of 60 runs. It was significantly better than DE and BB-BC on all trusses, but not better than SA or PSO on every truss. The exact solver was fastest on the 6-bar, needed 127 s on the 25-bar, and could not close a 48 % gap on the 72-bar in 90 min. These results define where exact methods are sufficient and where heuristics are still needed.

Keywords: Discrete truss optimization; Mixed-integer linear programming; Global optimality; Genetic algorithm; Local search; IS 800:2007; Benchmarking

Article Highlights

- Discrete IS 800 truss sizing is posed exactly as an MILP and solved to certified optima.
- A GA plus neighbourhood search returns designs with local-optimality certificates.
- Seven heuristics are compared at equal analysis budgets against certified optima.

1 Introduction

Space trusses are pin-jointed three-dimensional structures that carry load mainly by axial forces. They are used for long-span roofs, towers and industrial structures. In practice each member is chosen from a catalogue of rolled sections, so the minimum-weight design problem is discrete. In India the sections come from SP 6(1) [3] and the members must satisfy IS 800:2007 [2]: tension yielding, column buckling with the IS 800 buckling curve, and slenderness limits. Stiffness and displacement limits add further constraints.

Two research lines address this problem. Population-based metaheuristics have been tested for three decades on the same small set of space trusses, in particular the 25-bar and 72-bar benchmarks [28, 4, 5, 27, 7, 31]. In parallel, exact mixed-integer methods have solved discrete truss and frame problems to proven global optimality [1, 29, 16, 23, 36]. The two lines rarely meet. Stolpe [33] pointed out that, as a result, heuristic designs are compared only with other heuristic designs, methods are compared with different numbers of structural analyses, and many benchmarks that are routinely solved by heuristics could be solved exactly. Section 2 reviews both lines in more detail.

The present paper takes this critique as its starting point. We study discrete sizing of space trusses under IS 800:2007 and ask three questions. Can the complete IS 800 problem be solved exactly, and up to what size? How close do common metaheuristics come to the certified optimum when they all receive the same analysis

budget? Can a heuristic return a design together with a verifiable statement about its quality, instead of only a weight?

1.1 Contributions

1. We extend the exact MILP approach of Rasmussen and Stolpe [29], Mela [23] and Van Mellaert et al. [36] to IS 800:2007 and the SP 6(1) catalogue: tension, the IS 800 buckling curve, slenderness (including stress reversal between load cases) and displacement limits. To our knowledge this is the first certified global optimum for an IS 800 truss problem. We then use the certified optima, and a proved lower bound where the MILP does not finish, as the reference for judging the heuristics (Sections 4.1 and 7.1).

2. We propose GA–DNS: a genetic algorithm for exploration followed by a deterministic discrete neighbourhood search (DNS). DNS stops only when every neighbouring design has been analysed and is either heavier or infeasible, so each final design carries a local-optimality certificate (Section 4.2).

3. We compare GA–DNS with six metaheuristics and two deterministic methods under one protocol: the same analysis model, the same feasibility rule, the same analysis budget, 20 seeded runs, distances to certified optima, data profiles and corrected non-parametric tests (Sections 5 and 7).

4. All code, raw results and certificates are openly available.

We use "best found" for heuristic results and "optimal" only when a certificate exists. We also report where the proposed method is *not* better than simpler alternatives.

Section 2 reviews related work, Section 3 states the problem, Section 4 the methods, Section 5 the comparison protocol and Section 6 the benchmarks. Section 7 presents the results, Section 8 discusses them and Section 9 concludes.

2 Related work

2.1 Metaheuristics on benchmark trusses

Genetic algorithms with a penalty transformation were among the first methods applied to discrete structural sizing [28]. They were followed by ant colony optimization [4], particle swarm optimization [27], Big Bang–Big Crunch [5], simulated annealing [20], teaching–learning-based optimization [7] and many other population-based methods; Saka and Geem [30] review more than a hundred such applications to steel structures. Most of these studies use the same small benchmarks (10-, 25-, 72- and 120-bar trusses) and report the lightest design found. Three common weaknesses have been pointed out. First, the designs are compared only with other heuristic designs, so the distance from the true optimum is unknown [33]. Second, the methods are often compared with different numbers of structural analyses and with tuned parameters, which makes the ranking unreliable. Hasançebi et al. [13] compared seven metaheuristics on four real-size pin-jointed structures designed to ASD-AISC, on one common coding platform; simulated annealing and evolution strategies were the strongest, while the simple GA converged slowly. Third, many "new" algorithms differ from older ones mainly in their metaphor, not in their search mechanism [32]. Good practice therefore requires equal budgets, many independent runs and proper non-parametric statistics [8], and profiles that show performance as a function of budget [10, 24].

2.2 Exact methods for discrete truss and frame sizing

For linear-elastic trusses with a finite set of cross-sections, the discrete problem can be written as a mixed-integer program with binary section-choice variables and continuous displacements and forces. Achtziger and Stolpe [1] solved discrete truss topology problems to guaranteed global optimality by branch-and-bound and published benchmark examples with certified solutions. Rasmussen and Stolpe [29] added valid cuts and parallel branch-and-bound. Kanno and Guo [16] formulated robust truss topology design with stress constraints as an MILP. Mela [23] showed that member buckling, which is difficult in continuous truss optimization, becomes simple in a mixed-variable formulation, because the buckling resistance of each candidate

Table 1: Representative studies on discrete or catalogue-based truss and frame sizing. "Proof": global optimality is certified. "Equal NFE": all methods compared at the same number of structural analyses.

Study	Method	Structures	Proof	Equal NFE	Constraints / code
Rajeev and Krishnamoorthy [28]	GA	3-bar truss to 160-bar tower	–	–	penalty on violation
Camp and Bichon [4]	ACO	space trusses	–	–	stress, deflection; several load cases
Camp [5]	BB-BC	benchmark space trusses	–	–	stress, deflection
Hasançebi et al. [13]	7 metaheuristics	4 real-size pin-jointed	–	common platform	ASD-AISC
Degertekin and Hayalioglu [7]	TLBO	truss benchmarks	–	–	stress, displacement
Achtziger and Stolpe [1]	branch-and-bound	truss topology	✓	n/a	discrete areas
Rasmussen and Stolpe [29]	cut-and-branch	truss topology	✓	n/a	discrete areas
Mela [23]	MILP	truss topology	✓	n/a	member buckling
Van Mellaert et al. [36]	MILP vs. GA	steel frames	✓	–	Eurocode checks
This work	MILP + GA–DNS + 6 heuristics	space trusses	✓[a]	✓	IS 800:2007, SP 6(1)

[a] 6-bar and 25-bar certified; 72-bar with proved lower bound.

profile is a constant. Van Mellaert et al. [35] obtained global optima for statically determinate trusses with displacement, member and joint constraints, and Van Mellaert et al. [36] extended the MILP approach to discrete sizing of steel frames with Eurocode member checks, comparing it with a genetic algorithm. Stolpe [33] concluded that many benchmark problems solved by metaheuristics in the literature can be solved to global optimality by such methods. The limits of the exact approach are problem size (the number of binaries grows with groups $\times$ sections) and nonlinear behaviour that does not fit a linear model.

2.3 Hybrid and memetic algorithms

Combining a population-based search with a local improvement step is the idea of memetic algorithms [25]. Krasnogor and Smith [19] gave a design taxonomy and Neri and Cotta [26] a review of the field. In structural optimization, GA phases are often followed by gradient, pattern or local discrete searches. The usual purpose is faster convergence. A by-product that is seldom used is that a complete local search in a finite neighbourhood ends with a checkable statement: no neighbouring design is both feasible and lighter. GA–DNS in this paper is a memetic algorithm in this sense; its new element is that this certificate is reported for every run and compared with the exact global optimum.

2.4 Beyond linear elasticity

Where the structural response is nonlinear, for example with global stability or geometric nonlinearity, exact MILP formulations are no longer directly available [9, 23]. This is where heuristics remain most useful, and it is the motivation for measuring their quality carefully on problems where the exact answer is known.

2.5 Research gap

Table 1 summarises representative studies. Three gaps follow.

1. Heuristic results on space-truss benchmarks are rarely measured against a certified optimum or a proved lower bound, and rarely at equal analysis budgets.
2. Exact MILP methods have been applied with generic stress, displacement and buckling limits and with Eurocode checks, but not, to our knowledge, with IS 800:2007 and the SP 6(1) catalogue, which govern steel design in India.
3. Heuristics return a weight but no statement about the quality of the design; a local-optimality certificate is a cheap way to provide one.

3 Problem formulation

3.1 Design variables and objective

The geometry and the connectivity are fixed. The n_m members are divided into n_g design groups; all members of group g use the same section. The design variable of group g is an integer index $k_g \in \{1, \ldots, n_k\}$ into the

SP 6(1) catalogue of $n_k = 51$ equal angles, sorted by area. The weight is

$$W(\mathbf{k}) = \rho \sum_{g=1}^{n_g} A_{k_g} L_g, \qquad L_g = \sum_{i \in g} L_i, \tag{1}$$

with $\rho = 7850$ kg/m^3. Self-weight is not applied as a load, as in the original benchmarks.

3.2 Structural analysis

Let $\mathbf{B} \in \mathbb{R}^{n_m \times n_f}$ be the compatibility matrix that maps the free nodal displacements $\mathbf{u}$ to member elongations, $\mathbf{e} = \mathbf{B}\mathbf{u}$. For load case l the linear-elastic response follows from

$$\mathbf{K}(\mathbf{k})\,\mathbf{u}_l = \mathbf{p}_l, \qquad \mathbf{K} = \mathbf{B}^{\mathsf{T}}\,\mathrm{diag}\!\left(\frac{EA_i}{L_i}\right)\mathbf{B}, \qquad N_{il} = \frac{EA_i}{L_i}\,\mathbf{b}_i^{\mathsf{T}}\mathbf{u}_l, \tag{2}$$

with $E = 200$ GPa. $\mathbf{K}$ is factorised by Cholesky decomposition. Before any optimization, each model is checked for kinematic stability: the smallest eigenvalue of $\mathbf{K}$ (all members with the same section) must be positive. This simple check prevents a singular model from silently returning zero member forces.

3.3 IS 800:2007 member checks

Members are treated as concentrically loaded pin-ended bars with effective length KL, $K = 1$. For section k the design capacities are

$$T_k = \frac{A_k f_y}{\gamma_{m0}}, \tag{3}$$

$$C_{ik} = A_k f_{cd}(\lambda_{ik}), \quad \lambda_{ik} = \frac{KL_i}{r_k}, \quad f_{cd} = \frac{f_y/\gamma_{m0}}{\phi + \sqrt{\phi^2 - \bar{\lambda}^2}} \le \frac{f_y}{\gamma_{m0}}, \tag{4}$$

where $\bar{\lambda} = \sqrt{f_y/f_{cc}}$, $f_{cc} = \pi^2 E/\lambda^2$, $\phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$ and $\alpha = 0.49$ (IS 800 Cl. 7.1.2.1 with Table 7; buckling class c for angles, Table 10), with $f_y = 250$ MPa and $\gamma_{m0} = 1.10$. r_k is the minimum radius of gyration. A design is feasible if, for every member i and load case l,

$$-C_{ik_g} \le N_{il} \le T_{k_g}, \qquad \lambda_{ik_g} \le \begin{cases} 180 & \text{if } N_{il} < 0 \text{ in any load case,} \\ 400 & \text{otherwise,} \end{cases} \qquad |u_{dl}| \le \delta_{\lim} \;\; \forall d \in \mathcal{D}, \tag{5}$$

where $\mathcal{D}$ is the set of displacement-limited degrees of freedom. The slenderness limits follow IS 800 Table 3. The ratio of demand to capacity is reported as DCR.

4 Methods

4.1 Exact MILP

For fixed member i and section k, the quantities T_k, C_{ik} and λ_{ik} in Equations (3) to (5) are *constants*. The non-smooth buckling curve therefore does not enter the optimization as a function; it only fixes one number for each member–section pair. Following the displacement-based formulation of Rasmussen and Stolpe [29] and Stolpe [33], we introduce binary variables x_{gk} (group g uses section k), the displacements $\mathbf{u}_l$, and elongation

variables e_{ikl}, which carry the elongation of member i only for the selected section:

$$\min \rho \sum_g \sum_k A_k L_g \, x_{gk} \tag{6}$$

$$\text{s.t. } \textstyle\sum_k x_{gk} = 1 \qquad \forall g \tag{7}$$

$$\mathbf{b}_i^{\mathsf{T}} \mathbf{u}_l = \textstyle\sum_k e_{ikl} \qquad \forall i, l \tag{8}$$

$$\textstyle\sum_i \mathbf{b}_i \sum_k \frac{EA_k}{L_i} \, e_{ikl} = \mathbf{p}_l \qquad \forall l \tag{9}$$

$$-\frac{L_i C_{ik}}{EA_k} \, x_{gk} \le e_{ikl} \le \frac{L_i T_k}{EA_k} \, x_{gk} \qquad \forall i \in g,\ k,\ l \tag{10}$$

$$|u_{dl}| \le \delta_{\lim} \qquad \forall d \in \mathcal{D},\ l \tag{11}$$

$$x_{gk} \in \{0, 1\}.$$

If $x_{gk} = 0$, Equation (10) forces $e_{ikl} = 0$; if $x_{gk} = 1$, the member force is $N_{il} = (EA_k/L_i)e_{ikl}$ and Equation (10) is exactly the tension and buckling check. The same idea—one continuous force or elongation variable per candidate section—is used by Mela [23] and Van Mellaert et al. [36]; here it is applied to the IS 800 capacities. Slenderness is handled in the same way: if $\lambda_{ik} > 180$ we set $C_{ik} = 0$ (the member may not be in compression with that section), and if $\lambda_{ik} > 400$ for a member of group g we fix $x_{gk} = 0$. The model is exact, and it contains no big-M constants, which gives a much tighter continuous relaxation than the usual big-M form (in our tests the big-M version did not find any feasible design for the 25-bar truss in 150 s). Given an incumbent weight W^{inc}, sections that are too heavy even when all other groups use the lightest section are removed. The MILP is solved with the branch-and-bound solver of HiGHS [15], warm-started with the best heuristic design, a relative gap tolerance of 10^{-6} and one CPU core. Every returned design is re-analysed with the stiffness method of Equation (2).

4.2 Proposed GA–DNS hybrid

GA–DNS has two phases (Algorithm 1).

Phase 1 (exploration). An integer GA with 40 individuals, binary tournament selection, uniform crossover ($p_c = 0.9$), mutation rate $1/n_g$ (80 % ± 1 or ± 2 catalogue steps, 20 % random reset) and elitism uses 50 % of the analysis budget. The best feasible design and the three best distinct designs of the last population are passed to Phase 2.

Phase 2 (deterministic refinement with certificate). Each elite is first made feasible, if necessary, by a greedy repair that moves the group with the largest constraint ratio one catalogue step up (for displacement violations, the step with the best stiffness gain per kilogram is chosen). The DNS then performs steepest descent in the neighbourhood

$$\mathcal{N}(\mathbf{k}) = \{\mathbf{k} - \mathbf{e}_i\} \ \cup \ \{\mathbf{k} - \mathbf{e}_i + \mathbf{e}_j : \ i \neq j, \ W(\mathbf{k} - \mathbf{e}_i + \mathbf{e}_j) < W(\mathbf{k})\}, \tag{12}$$

i.e. one group one step lighter, or one group lighter and another group heavier (a "swap"). Because the catalogue is sorted by area, every $\mathbf{k} - \mathbf{e}_i$ is lighter. The lighter neighbours are analysed in order of increasing weight, and the first feasible one is accepted. When no neighbour is feasible and lighter, DNS stops. At that point every design in $\mathcal{N}(\mathbf{k})$ has been analysed, so the final design is *certified* to be a local minimum with respect to $\mathcal{N}$. The certificate is a finite, checkable statement; it is not a proof of global optimality. With the remaining budget, an iterated local search [21] changes two random groups of the best certified design by up to three catalogue steps and runs DNS again; a new design is kept only if it is certified and lighter.

4.3 Comparison methods

Six metaheuristics act on the same integer variables: simulated annealing (SA) [18], the same GA as in Phase 1 run for the full budget, particle swarm optimization (PSO) [17, 27], ant colony optimization (ACO) [11, 4], Big Bang–Big Crunch (BB-BC) [12, 5] and differential evolution (DE) [34]. Continuous positions (PSO,

Algorithm 1 GA–DNS

Require: model, analysis budget B

Run the GA (Phase 1) until $0.5B$ analyses are used
$\mathcal{E} \leftarrow$ best feasible design and 3 best distinct designs of the last population
for each $\mathbf{k} \in \mathcal{E}$ **do**
    $\mathbf{k} \leftarrow \textsc{Repair}(\mathbf{k})$; $\mathbf{k} \leftarrow \mathrm{DNS}(\mathbf{k})$ ▷ certified local minimum
end for
$\mathbf{k}^\star \leftarrow$ lightest certified design
while analyses $< B$ **do** ▷ iterated local search
    $\mathbf{k}' \leftarrow \mathbf{k}^\star$ with two random groups changed by ≤ 3 steps
    $\mathbf{k}' \leftarrow \textsc{Repair}(\mathbf{k}')$; $\mathbf{k}' \leftarrow \mathrm{DNS}(\mathbf{k}')$
    if $\mathbf{k}'$ is certified and $W(\mathbf{k}') < W(\mathbf{k}^\star)$ **then** $\mathbf{k}^\star \leftarrow \mathbf{k}'$
end while
return $\mathbf{k}^\star$ and its certificate

function DNS($\mathbf{k}$)
    repeat
        analyse the members of $\mathcal{N}(\mathbf{k})$ that are lighter than $\mathbf{k}$, lightest first
        move to the first feasible one; if none exists, **return** $\mathbf{k}$ (certified)
    until budget exhausted
end function

Table 2: Method settings (identical for all trusses).

Method	Settings
SA	one group changed by ±1–3 steps; T from $0.2W_{\max}$ to $10^{-4}W_{\max}$, geometric
GA	population 40; tournament 2; uniform crossover 0.9; mutation $1/n_g$; elitism 1
PSO	30 particles; inertia $0.9 \to 0.4$; $c_1 = c_2 = 2$; $\lvert v\rvert \leq 10$
ACO	20 ants; $\alpha = 1$, $\beta = 0.5$ ($\eta = 1/A_k$); evaporation 0.2; rank-based (3) + elitist update
BB-BC	30 candidates; centre = 0.5 (centre of mass)+0.5 (best); spread $\propto 1/t$
DE	DE/rand/1/bin; population 30; $F = 0.6$; $CR = 0.9$
GA–DNS	GA as above for $0.5B$; 4 elites; DNS neighbourhood Equation (12); ILS kick 2 groups, ≤ 3 steps
FSD	start at mid-catalogue; at most 50 resizing cycles
SQP-R	SLSQP, 200 iterations, ftol = 10^{-9}; round up; repair
MILP	HiGHS 1.15, gap 10^{-6}, 1 thread, time limit 5400 s, warm start

BB-BC, DE) are rounded to the nearest index before analysis. Two deterministic methods are included: fully stressed design with catalogue selection (FSD), which cannot control displacements, and SQP-R, a continuous SLSQP solution [38] with radius of gyration from a fitted law $r = c\,A^b$, followed by rounding up and the repair step of GA–DNS. Table 2 lists all settings. The parameters are common textbook values and were not tuned for any benchmark.

5 Comparison protocol

Same analysis budget. One analysis (NFE) is one solution of Equation (2) for all load cases of one design. Every heuristic stops after $B = 3000$, 10 000 and 20 000 analyses for the 6-, 25- and 72-bar trusses. Repeated analyses of the same design are counted.

Same constraint handling. All heuristics rank designs by the feasibility rule of Deb [6], written as one scalar: $f = W$ for feasible designs and $f = W_{\max} + 10^3 v$ otherwise, where $W_{\max}$ is the weight of the heaviest catalogue design and v is the sum of constraint violations (normalised ratios minus one). A feasible design is therefore always better than an infeasible one, and no penalty factor needs tuning. All reported designs are re-checked by an independent code path (Section 7.5).

Table 3: Benchmark definitions.

Truss	Nodes	Members	Groups	Load cases	Budget B	Displacement limit
6-bar	4	6	6	1	3000	none
25-bar	10	25	8	1	10 000	8.89 mm, all nodes
72-bar	20	72	16	2	20 000	6.35 mm, top nodes x, y

Runs and statistics. Each stochastic method is run with seeds 0–19. We report best, median, mean, standard deviation, the share of runs that reach the certified (or best-known) weight W^*, and the median number of analyses used to find the final design. Because the runs of different methods are independent, GA–DNS is compared with each competitor by the one-sided Mann–Whitney U test [22] with Holm correction [14], and the effect size is the Vargha–Delaney A_{12} [37]. Convergence is plotted against analyses, not against time or iterations, and data profiles [24] show the share of all runs that reach W^* (or $1.01W^*$) as a function of the budget used.

Software. Python 3.13, NumPy, SciPy [38] and HiGHS [15], on a 2-core Linux machine.

6 Benchmark trusses

Figure 1 and Table 3 summarise the three trusses.

6-bar tetrahedron. Four nodes, six members, each member its own group. The apex carries 50 kN in y and -100 kN in z. No displacement limit. This small case is used to check the implementation.

25-bar transmission tower. The ASCE 25-bar tower with 10 nodes, 25 members in 8 groups and supports at the four base nodes, with the geometry and load case used by Camp and Bichon [4]: node 1: $(1, -10, -10)$ kip; node 2: $(0, -10, -10)$ kip; nodes 3 and 6: 0.5 kip in x. All nodal displacements are limited to ± 0.35 in (8.89 mm).

72-bar four-storey truss. Twenty nodes, 72 members in 16 groups (per storey: verticals, face diagonals, horizontals, plan diagonals), plan 120×120 in, storey height 60 in, supports at the four base nodes, and two load cases [5]: LC1, one top node $(5, 5, -5)$ kip; LC2, all four top nodes $(0, 0, -5)$ kip. The x and y displacements of the top nodes are limited to ± 0.25 in (6.35 mm).

The benchmark geometries and loads are taken from the earlier literature (imperial units, US sections and allowable-stress limits), but the section catalogue and the member checks here are those of IS 800 and SP 6(1). Therefore the weights in this paper must not be compared directly with the published results.

7 Results

7.1 Certified optima

Table 4 gives the exact MILP results. For the 6-bar truss the MILP proves in 0.02 s that 101.17 kg is the global optimum. For the 25-bar truss it proves that 580.06 kg is the global optimum after 6753 branch-and-bound nodes and 127 s. For the 72-bar truss the solver reached the 90-minute limit with the warm-start design of 1032.47 kg as the upper bound and a proved lower bound of 535.17 kg. The 72-bar optimum is therefore only known to lie in [535.17, 1032.47] kg; we call 1032.47 kg the best-known value. This shows the practical limit of the exact approach on a laptop-class machine: 16 groups $\times$ 51 sections and two load cases are already difficult.

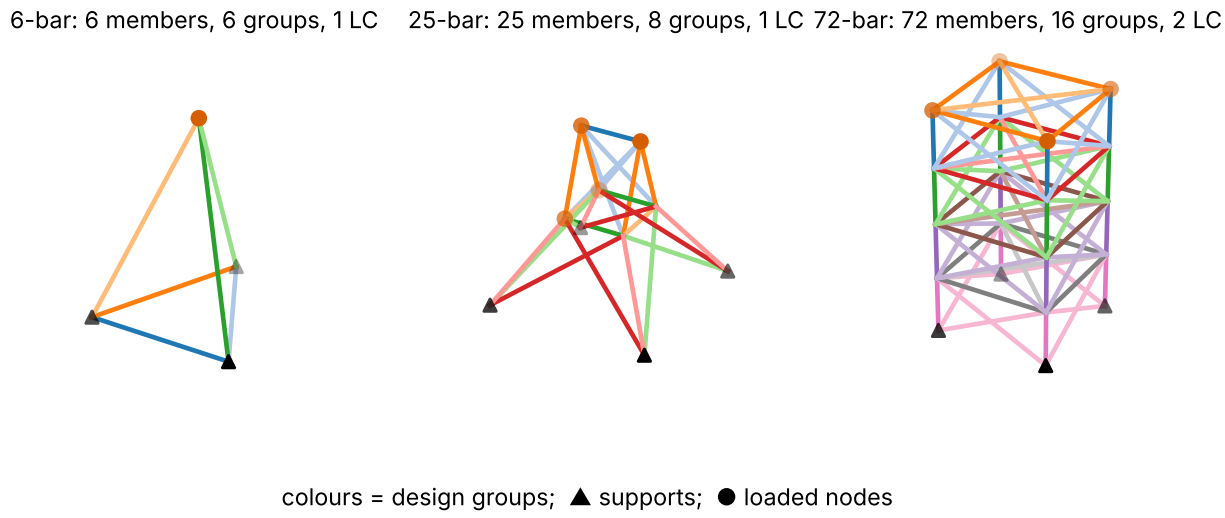


Figure 1: Benchmark trusses. Colours show design groups; triangles are supports and orange dots are loaded nodes.

Table 4: Exact MILP (Section 4.1) solved with HiGHS branch-and-bound on one CPU core. Upper bound = best feasible design; lower bound = proved bound on the global optimum. The 72-bar run was stopped at the 5400 s time limit.

Truss	Binary	Continuous	Rows	B&B nodes	Time (s)	Upper (kg)	Lower (kg)	Status
6-bar	306	312	476	0	0.0	101.17	101.17	optimal (gap 0)
25-bar	408	1,293	2,257	6,753	126.8	580.06	580.06	optimal (gap 0)
72-bar	816	7,440	13,440	6,680	5400.1	1032.47	535.17	time limit (gap 48.2 %)

Table 5: Results over $N = 20$ independent runs (seeds 0–19) with an equal budget of structural analyses for every method. All final designs are IS 800:2007-feasible (success rate 100 % for every method). Hit = share of runs that reach W^* exactly; NFE = median number of analyses needed to find the final best design. Bold = equal to W^*. FSD and SQP-R are deterministic (one run).

Method	Best (kg)	Median (kg)	Mean (kg)	Std (kg)	Hit (%)	NFE	$\bar{t}$ (s)
6-bar — budget 3,000 analyses; $W^* = 101.17$ kg (certified global optimum)							
GA–DNS*	**101.17**	**101.17**	101.23	0.27	95	1,920	0.25
GA	**101.17**	**101.17**	101.90	0.91	55	2,508	0.25
DE	103.64	105.25	105.37	1.06	0	2,769	0.33
PSO	**101.17**	102.39	102.29	0.79	25	2,714	0.22
SA	**101.17**	**101.17**	101.17	0.00	100	2,354	0.33
ACO	**101.17**	**101.17**	102.46	2.31	60	2,046	0.36
BB-BC	102.59	107.95	108.95	5.11	0	1,751	0.23
FSD	**101.17**	—	—	—	100	2	0.00
SQP-R	107.87	—	—	—	0	84	0.01
25-bar — budget 10,000 analyses; $W^* = 580.06$ kg (certified global optimum)							
GA–DNS*	**580.06**	**580.06**	583.87	16.36	90	5,033	1.01
GA	**580.06**	**580.06**	583.91	16.54	90	6,392	1.01
DE	**580.06**	**580.06**	589.12	20.82	70	6,322	1.33
PSO	**580.06**	**580.06**	580.35	0.89	90	6,335	0.95
SA	**580.06**	**580.06**	581.36	1.47	55	8,282	1.20
ACO	**580.06**	583.87	587.00	12.61	20	3,572	1.58
BB-BC	613.32	646.53	661.94	42.18	0	2,500	0.95
FSD	584.80	—	—	—	0	4	0.01
SQP-R	602.77	—	—	—	0	104	0.01
72-bar — budget 20,000 analyses; $W^* = 1032.47$ kg (best known; not certified)							
GA–DNS*	**1032.47**	1059.45	1056.21	16.53	10	16,127	2.64
GA	1033.62	1064.62	1062.27	17.78	0	17,401	2.70
DE	1075.36	1103.03	1112.38	30.20	0	19,316	3.41
PSO	1040.60	1066.13	1069.92	19.27	0	19,534	2.81
SA	**1032.47**	1034.44	1045.69	23.76	25	19,522	3.11
ACO	1033.62	1082.87	1098.83	71.08	0	8,175	5.06
BB-BC	1341.92	1522.71	1531.33	126.56	0	3,678	2.62
FSD	1803.03	—	—	—	0	1	0.03
SQP-R	1094.93	—	—	—	0	318	0.06

7.2 Statistical comparison

Table 5 lists the results of all methods. Every method produced a feasible design in every run, because of the common feasibility rule.

- **6-bar.** GA–DNS reached the global optimum in 19 of 20 runs and SA in all 20 runs. FSD also finds the optimum, using only two analyses, because this truss is statically determinate and has no displacement limit. DE and BB-BC never reached the optimum.
- **25-bar.** GA–DNS, GA and PSO reached the certified optimum in 90 % of the runs; PSO had the lowest mean (580.35 kg). The two GA–DNS runs that missed the optimum stopped in other certified local minima (582.94 kg and 653.33 kg).
- **72-bar.** SA and GA–DNS found the best-known design (1032.47 kg), in 25 % and 10 % of the runs respectively. GA–DNS had a smaller spread (std 16.5 kg vs 23.8 kg) and a better worst case, but SA had the better median (1034.44 kg vs 1059.45 kg).

Table 6 confirms these observations. After Holm correction, GA–DNS gives significantly lighter designs than DE and BB-BC on all trusses, than ACO on the 6- and 25-bar trusses, than GA and PSO on the 6-bar

Table 6: One-sided Mann–Whitney U tests (H_1: GA–DNS gives lighter final designs), Holm-corrected within each truss, and Vargha–Delaney effect size A_{12} (probability that a GA–DNS run is lighter than a competitor run; 0.5 = no difference).

Truss	Competitor	p_{Holm}	A_{12}	Conclusion
6-bar	GA	0.005	0.71	GA–DNS lighter
	DE	<0.001	1.00	GA–DNS lighter
	PSO	<0.001	0.86	GA–DNS lighter
	SA	0.853	0.47	n.s.; competitor better
	ACO	0.007	0.68	GA–DNS lighter
	BB-BC	<0.001	1.00	GA–DNS lighter
25-bar	GA	1.000	0.50	n.s.
	DE	0.189	0.60	n.s.
	PSO	1.000	0.50	n.s.; competitor better
	SA	0.047	0.66	GA–DNS lighter
	ACO	<0.001	0.84	GA–DNS lighter
	BB-BC	<0.001	0.97	GA–DNS lighter
72-bar	GA	0.273	0.60	n.s.
	DE	<0.001	0.98	GA–DNS lighter
	PSO	0.055	0.70	n.s.
	SA	0.992	0.28	n.s.; competitor better
	ACO	0.129	0.66	n.s.
	BB-BC	<0.001	1.00	GA–DNS lighter

truss, and than SA on the 25-bar truss. It is not significantly better than GA or PSO on the larger trusses, nor than SA on the 6- and 72-bar trusses; on the 72-bar truss the effect size even favours SA ($A_{12} = 0.28$).

7.3 Convergence and data profiles

Figure 2 shows the median best-feasible weight against the number of analyses. ACO falls fastest at the start but then stagnates. On the 6- and 25-bar trusses, GA–DNS reaches W^* earlier than the other methods; the step at $0.5B$ marks the start of DNS. On the 72-bar truss SA is slow at first (high temperature) but reaches the lowest median at the end of the budget. The data profiles in Figure 3 combine all 60 runs per method. For the exact target W^*, GA–DNS solves the largest share of runs for every budget fraction above about 0.55, ending at 65 % of all runs (SA 60 %, GA 48 %). For the 1 % target, GA–DNS leads between about 0.55 and 0.8 of the budget, but at the full budget SA solves the most runs (93 % against 72 % for GA–DNS), mainly because of its strong 72-bar results. The distributions of W/W^* are shown in Figure 4.

7.4 Effect of Phase 2 and the certificates

Compared with the best design at the end of Phase 1, the DNS and ILS steps reduced the final weight by a median of 2.8 % (6-bar), 0.2 % (25-bar) and 2.4 % (72-bar), with a maximum of 9.8 %, 6.6 % and 6.7 %. The final design of GA–DNS was certified as a local minimum with respect to Equation (12) in 20/20, 20/20 and 19/20 runs; in the remaining 72-bar run the best design was found at analysis 19 945 of 20 000, and the budget ended before DNS could complete its check. On the 6- and 25-bar trusses, every run that did not reach W^* still returned a certified local minimum. This tells the user that the design cannot be improved by any single-step or swap change, and that a different starting point (or the MILP) is needed to improve it.

7.5 Designs and independent verification

Tables 7 to 9 give the best designs. Each was re-analysed by a separate implementation (element-by-element assembly, dense solver, IS 800 formulas coded again). The weights agree to 10^{-6} kg, the equilibrium error is below 10^{-14} relative to the load, and all constraints are satisfied: the largest DCR is 0.935, 0.998 and 0.881, and the largest displacement ratio is 0.524 (25-bar) and 0.353 (72-bar). In the 25-bar design the strength of group 3 (DCR 0.998) and the slenderness of group 5 (λ-ratio 0.994) are active, while the displacement limit

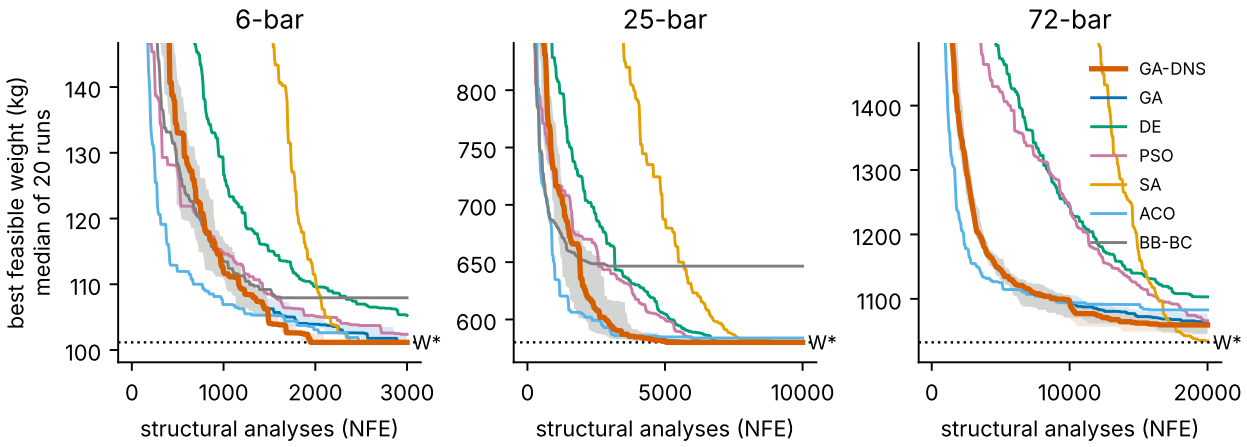


Figure 2: Median best-feasible weight of 20 runs against the number of structural analyses (shaded: interquartile range of GA–DNS and GA). Dotted line: certified optimum (6-, 25-bar) or best-known weight (72-bar).

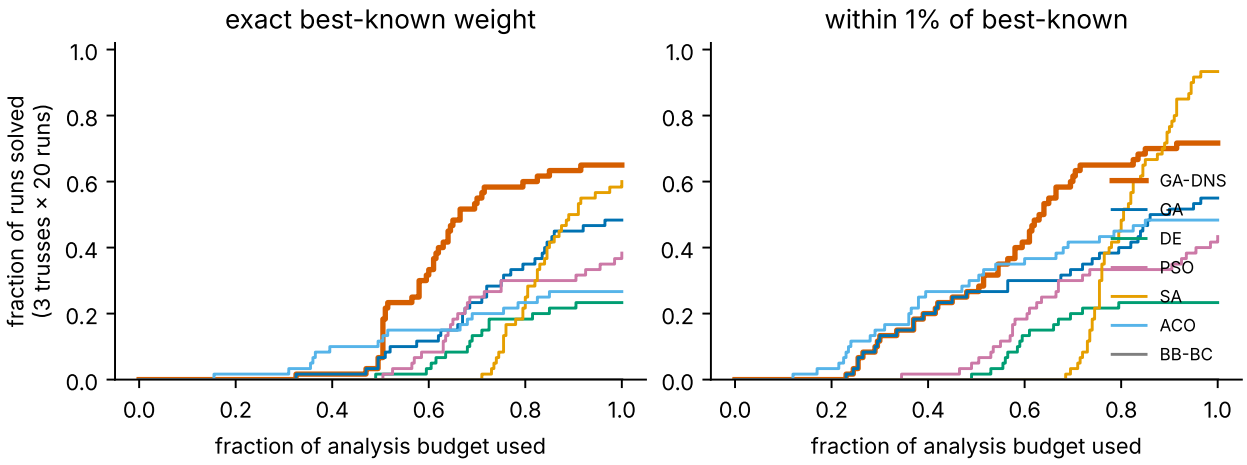


Figure 3: Data profiles over all three trusses (60 runs per method): share of runs that reach W^* exactly (left) or within 1 % (right) as a function of the fraction of the analysis budget used.

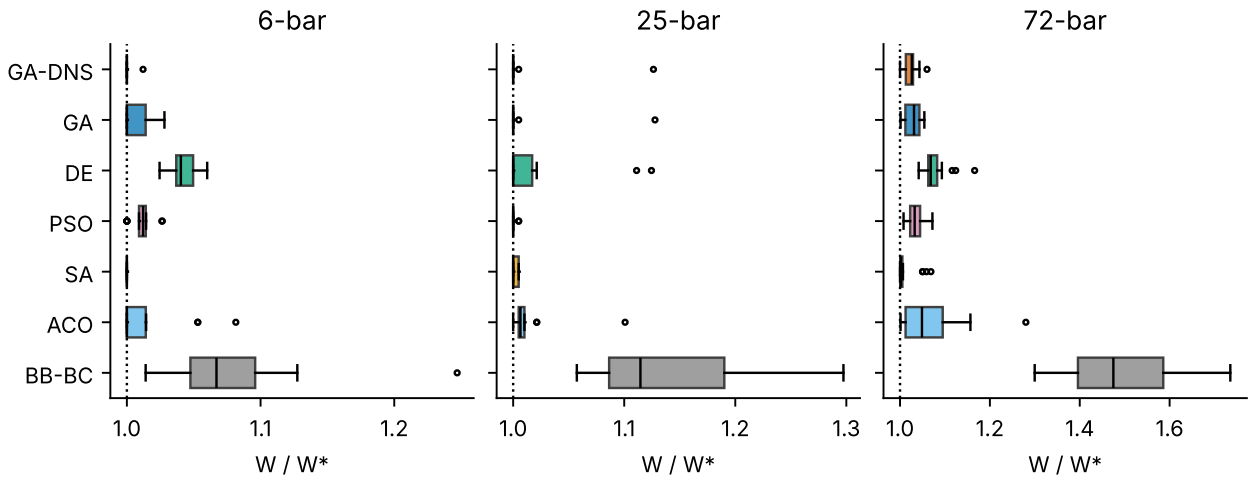


Figure 4: Distribution of final weight ratio W/W^* over 20 runs (1.0 = certified or best-known optimum).

Table 7: Certified optimal 6-bar design (one member per group). Weight 101.17 kg; max. displacement ratio 0.000. F = range of member forces over all load cases (negative = compression); DCR = max. demand/capacity (stress or Annex D buckling); λ-ratio = (KL/r)/limit.

Grp	n	ISA section	A (cm^2)	$F_{\min}$ (kN)	$F_{\max}$ (kN)	KL/r	DCR	λ-ratio
1	1	55×55×5	5.30	-14.1	-14.1	178	0.596	0.989
2	1	30×30×3	1.73	24.5	24.5	364	0.622	0.909
3	1	30×30×3	1.73	24.5	24.5	364	0.622	0.909
4	1	40×40×4	3.08	9.4	9.4	370	0.135	0.924
5	1	40×40×4	3.08	9.4	9.4	370	0.135	0.924
6	1	110×110×8	17.20	-124.6	-124.6	126	0.935	0.702

is not. In the 72-bar design group 7 is exactly at the tension slenderness limit of 400, and several compression groups are close to 180 (λ-ratio 0.92–0.97). Thus, with IS 800 slenderness limits and small SP 6(1) angles, slenderness rather than displacement governs these benchmarks under IS 800; in the original versions of the problems the displacement limits are usually reported as active.

8 Discussion

8.1 When is a heuristic justified?

For linear-elastic trusses the IS 800 problem is an exact MILP. On the 6-bar truss the MILP is the best method by any measure. On the 25-bar truss the MILP needs about 127 s, while GA–DNS finds the same design in about 1 s in 90 % of the runs; however, only the MILP proves that it is optimal. On the 72-bar truss the MILP could not close the gap in 90 minutes, so heuristics are still useful there, and better MILP techniques (valid cuts, symmetry breaking, parallel branch-and-bound [29]) are an obvious next step. Our answer to the critique of Stolpe [33] is therefore: heuristic results on these benchmarks should be measured against exact bounds, and a heuristic is justified mainly where the exact model stops being an MILP—for example with geometric nonlinearity, imperfection-sensitive buckling or nonlinear code interaction checks—or where it becomes too large.

8.2 What the certificate adds

A weight alone does not tell the engineer whether the design can still be improved. The DNS certificate is weaker than global optimality, but it is cheap (at most n_g^2 analyses per check), it can be verified by anyone,

Table 8: Certified optimal 25-bar design. Weight 580.06 kg; max. displacement ratio 0.524. F = range of member forces over all load cases (negative = compression); DCR = max. demand/capacity (stress or Annex D buckling); λ-ratio = (KL/r)/limit.

Grp	n	ISA section	A (cm^2)	$F_{\min}$ (kN)	$F_{\max}$ (kN)	KL/r	DCR	λ-ratio
1	1	25×25×3	1.42	5.5	5.5	250	0.169	0.625
2	4	65×65×6	7.52	-33.4	20.1	167	0.895	0.927
3	4	70×70×6	8.13	-62.9	26.3	126	0.998	0.702
4	2	20×20×3	1.11	1.1	1.9	313	0.074	0.782
5	2	35×35×4	2.67	-4.3	3.7	179	0.362	0.994
6	4	90×90×6	10.60	-20.8	8.1	165	0.389	0.918
7	4	90×90×6	10.60	-34.5	14.7	165	0.644	0.918
8	4	90×90×6	10.60	-72.5	37.1	122	0.835	0.676

Table 9: Best known 72-bar design (groups 1–4 = top storey; in each storey: verticals, face diagonals, horizontals, plan diagonals). Weight 1032.47 kg; max. displacement ratio 0.353. F = range of member forces over all load cases (negative = compression); DCR = max. demand/capacity (stress or Annex D buckling); λ-ratio = (KL/r)/limit.

Grp	n	ISA section	A (cm^2)	$F_{\min}$ (kN)	$F_{\max}$ (kN)	KL/r	DCR	λ-ratio
1	4	40×40×4	3.08	-18.4	-0.6	124	0.752	0.691
2	8	65×65×6	7.52	-12.2	5.1	172	0.343	0.953
3	4	60×60×5	5.82	-5.8	2.0	165	0.197	0.917
4	2	80×80×6	9.35	-7.8	2.9	175	0.182	0.971
5	4	40×40×4	3.08	-19.3	-1.1	124	0.788	0.691
6	8	65×65×6	7.52	-7.5	7.4	172	0.210	0.953
7	4	25×25×3	1.42	0.0	1.8	400	0.054	1.000
8	2	60×60×5	5.82	0.1	7.2	234	0.054	0.584
9	4	40×40×4	3.08	-20.5	11.3	124	0.837	0.691
10	8	65×65×6	7.52	-8.0	6.3	172	0.224	0.953
11	4	60×60×5	5.82	-0.9	4.5	165	0.034	0.917
12	2	45×45×4	3.49	0.0	2.7	311	0.034	0.779
13	4	45×45×4	3.49	-29.0	20.5	110	0.881	0.612
14	8	65×65×6	7.52	-10.7	8.4	172	0.301	0.953
15	4	60×60×5	5.82	-2.4	5.7	165	0.081	0.917
16	2	40×40×4	3.08	0.6	3.0	352	0.043	0.880

and it is independent of the random seed. Combined with the MILP lower bound, the user also gets a bound on the remaining gap. For the 72-bar truss, for example, the best GA–DNS design is certified locally optimal and lies at most $(1032.47 - 535.17)/1032.47 = 48\,\%$ above the global optimum. This bound is loose; it shows the value of a better lower bound rather than the quality of the design.

8.3 GA–DNS compared with the other heuristics

GA–DNS solved the largest share of runs to the exact target (Figure 3), and it reached W^* with the fewest analyses on the 6- and 25-bar trusses (median 1933 and 5033 analyses, against 2354 and 6466 for the next best method). It was not uniformly the best: at the 1 % target and the full budget SA was more reliable, mainly on the 72-bar truss. Well-tuned SA and PSO are strong on these small discrete problems, which agrees with the strong performance of SA reported by Hasançebi et al. [13], and the present paper does not claim otherwise. The main weakness of GA–DNS is that Phase 2 cannot leave a poor basin chosen by Phase 1 (the two 653 kg runs on the 25-bar truss); larger ILS kicks or several independent GA restarts may help.

8.4 Deterministic baselines

FSD needs very few analyses and is optimal for the statically determinate 6-bar truss, but it is 0.8 % heavier on the 25-bar and 75 % heavier on the 72-bar truss, because it ignores stiffness redistribution. Continuous SQP with rounding (SQP-R) is 6.6 %, 3.9 % and 6.0 % heavier than W^*; the rounding step and the approximate $r(A)$ law cost weight. Both are useful only as quick starting designs.

8.5 Limitations

- The analysis is linear-elastic with ideal pin joints. Single angles connected by one leg should be checked with the equivalent slenderness of IS 800 Cl. 7.5, and net-section rupture and block shear (Cl. 6.3–6.4) are not checked. These checks would change the capacities but not the MILP structure, as long as each capacity is a constant for a member–section pair.
- The loads come from the imperial benchmarks of the earlier literature and are not IS 875 load combinations.
- The method parameters were not tuned. Other settings could change the ranking of the heuristics, but they would not change the certified optima.
- Only one machine and one MILP solver were used; commercial solvers may close the 72-bar gap faster.

9 Conclusions

1. For linear-elastic space trusses with a fixed catalogue, the complete IS 800:2007 sizing problem (tension, column buckling, slenderness and displacements) can be written as an exact MILP, as earlier shown for other codes. The disaggregated formulation certified the global optima of the 6-bar (101.17 kg) and 25-bar (580.06 kg) trusses and gave a lower bound of 535.17 kg for the 72-bar truss.
2. GA–DNS, a GA followed by a deterministic neighbourhood search, returned a certified local minimum in 59 of 60 runs and reached the certified optimum in 95 % (6-bar) and 90 % (25-bar) of the runs at equal analysis budgets.
3. GA–DNS was significantly better than DE and BB-BC on all trusses and than ACO on two trusses, but it was not better than SA or PSO on every truss; SA had the best median on the 72-bar truss.
4. With IS 800 slenderness limits and SP 6(1) angles, slenderness and strength, not displacement, govern the optimal designs of these benchmarks.
5. Future work will use the same certified protocol for geometrically nonlinear and imperfection-sensitive space trusses, where an exact MILP is no longer available and heuristics with certificates are most needed.

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